

UrbanMind AI

Complete Project Workflow

From Data Collection to Urban Intelligence & Recommendations

Each stage: Tool • Importance • How It Is Done • Result

Data Engineering + Spatial Analytics + BI + Machine Learning + Explainable AI

1 — Data Collection

1 Data Collection

Tools: Python + Requests + APIs / CSV / Excel / GeoJSON + geographic sources such as OpenStreetMap

Importance: Build the raw data foundation representing the city from multiple dimensions.

How: Download/query sources, keep each dataset in its original format, and record source/date.

What we collect: Population, roads, traffic, public transportation, air quality, and green spaces.

Result: Separate raw files for each source, without modifying the original data.

2 — Spatial Integration

2 Geospatial Processing + H3

Tools: GeoPandas + OSMnx + H3 + Shapely

Importance: Put different datasets on the same spatial unit instead of analyzing each source separately.

H3: Divides the city into consistent spatial cells; each cell becomes a common analytical unit.

How: Process geometries, standardize CRS, then assign each feature to an H3 cell.

Result: Each H3 cell contains population, roads, traffic, transit, environmental, and green-space features.

Principle: One Spatial Unit → Multiple Urban Dimensions → One Decision Layer.

3 — SQLite Database

3 Integrated Database

Tools: SQLite + SQL + SQLAlchemy (optional for Python access)

Importance: Make SQLite the Source of Truth instead of relying on separate files.

How: Load integrated data, define Primary/Foreign Keys, and index important fields.

Design: H3_Features for the main analytical data + Dim_District for districts and relationships.

Result: One database from which any analytical or ML dataset can be extracted without recollecting data.

Current project design: 3,489 H3 cells and 41 districts.

4 — Preprocessing & Cleaning

4 Data Quality & Preprocessing

Tools: Pandas + NumPy + GeoPandas + SQL

Importance: Prevent data-quality problems from affecting analysis and models.

How: Treat/remove values according to their nature, standardize units, convert types, and validate relationships.

Checks: Missing values, duplicates, data types, outliers, invalid values, and spatial consistency.

Result: A Clean Analytical Dataset while preserving the original SQLite source.

5 — EDA & Spatial Analysis

5 Exploratory Data Analysis

Tools: Pandas + NumPy + Matplotlib + Seaborn + GeoPandas + PySAL when needed

Importance: Understand relationships and patterns before building models.

How: Descriptive statistics, correlation, distributions, maps, and spatial-cluster analysis when needed.

Analysis: Population↔Traffic, Population↔Transit, Traffic↔Air Quality, Population↔Green Space.

Result: Insights, candidate ML features, and identification of high-pressure or underserved areas.

6 — Power BI Decision-Support Layer

6 Power BI Analytics

Tools: Microsoft Power BI + DAX

Importance: Turn data into indicators and maps that explain the current state of the city.

Pages: City Overview, Urban Pressure, Mobility, Environmental, H3 Spatial Map.

How: Import the clean dataset, build KPIs and measures, and add filters and H3 maps.

Result: Interactive dashboards showing current conditions and spatial relationships.

7 — ML Preparation

7 Feature Engineering

Tools: Python + Pandas + NumPy + Scikit-learn

Importance: Transform raw indicators into features that better represent urban pressure.

How: Derive ratios and indicators, scale when needed, then split data for training/testing.

Examples: Population Density, Road Density, Intersection Density, Network Pressure, Transit Accessibility, NO₂, NDVI, Green-Space Coverage.

Result: An ML-ready dataset suitable for training and evaluation.

8 — Machine Learning

8 Models

Tools: Scikit-learn + XGBoost

Clustering: K-Means or HDBSCAN to discover area patterns without a target.

Comparison: Random Forest Regressor, then XGBoost Regressor as the proposed main model.

Baseline: Linear Regression or another simple model for comparison.

Evaluation: MAE + RMSE + R² for regression, with cross-validation when appropriate.

Goal: Estimate Urban Pressure / Vulnerability rather than using ML as a prediction exercise only.

9 — Explainable AI

9 SHAP

Tool: SHAP

Importance: Explain why each area received a specific prediction or score.

Example: High Population Density + High Traffic Pressure + Low Transit Access can increase intervention priority.

How: Measure the contribution of each feature to model outputs, locally and globally.

Result: A prediction that is understandable and defensible to users and decision-makers.

10 — Recommendation Engine

10 Urban Recommendations

Tools: Python Rule Engine + ML/SHAP outputs

Importance: Convert predictions into suggested actions instead of stopping at a number.

Example: High Traffic + High NO₂ → traffic-management priority; Low Transit + High Population → improve transit coverage.

How: Map problem patterns to clear recommendation rules.

Result: Analyze → Predict → Explain → Recommend.

11 — Final Interface

11 Delivery Layer

Tools: Power BI + Streamlit when an additional interactive application is needed

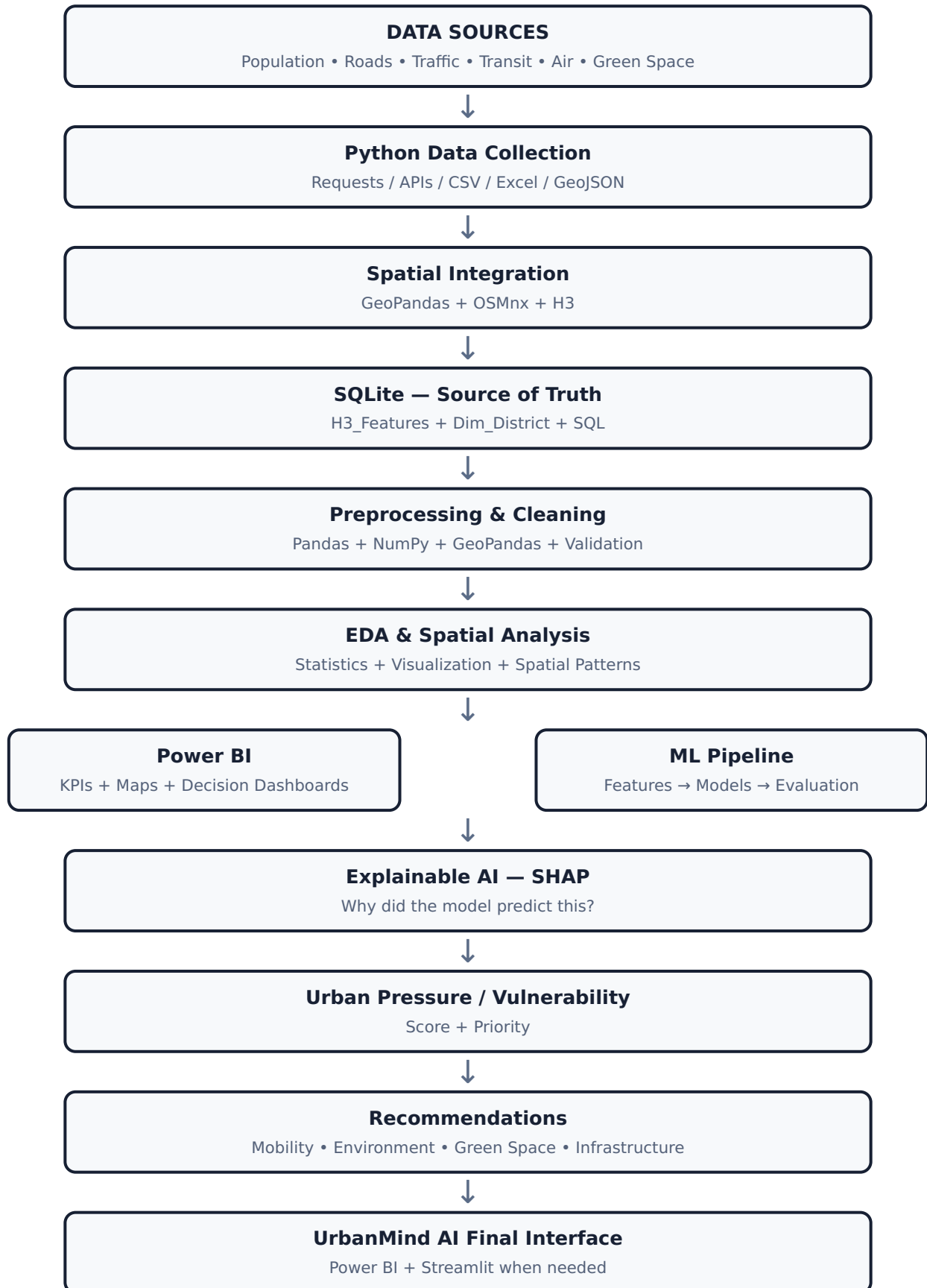
Importance: Bring the results together into one usable experience.

Result: An analytical Digital Twin interface connecting city conditions to predictions and actions.

How: Present maps, indicators, predictions, explanations, and recommendations.

Deliverables: Dashboard + ML outputs + Documentation + Database.

12 — Final Project Workflow



UrbanMind AI — End-to-End Workflow